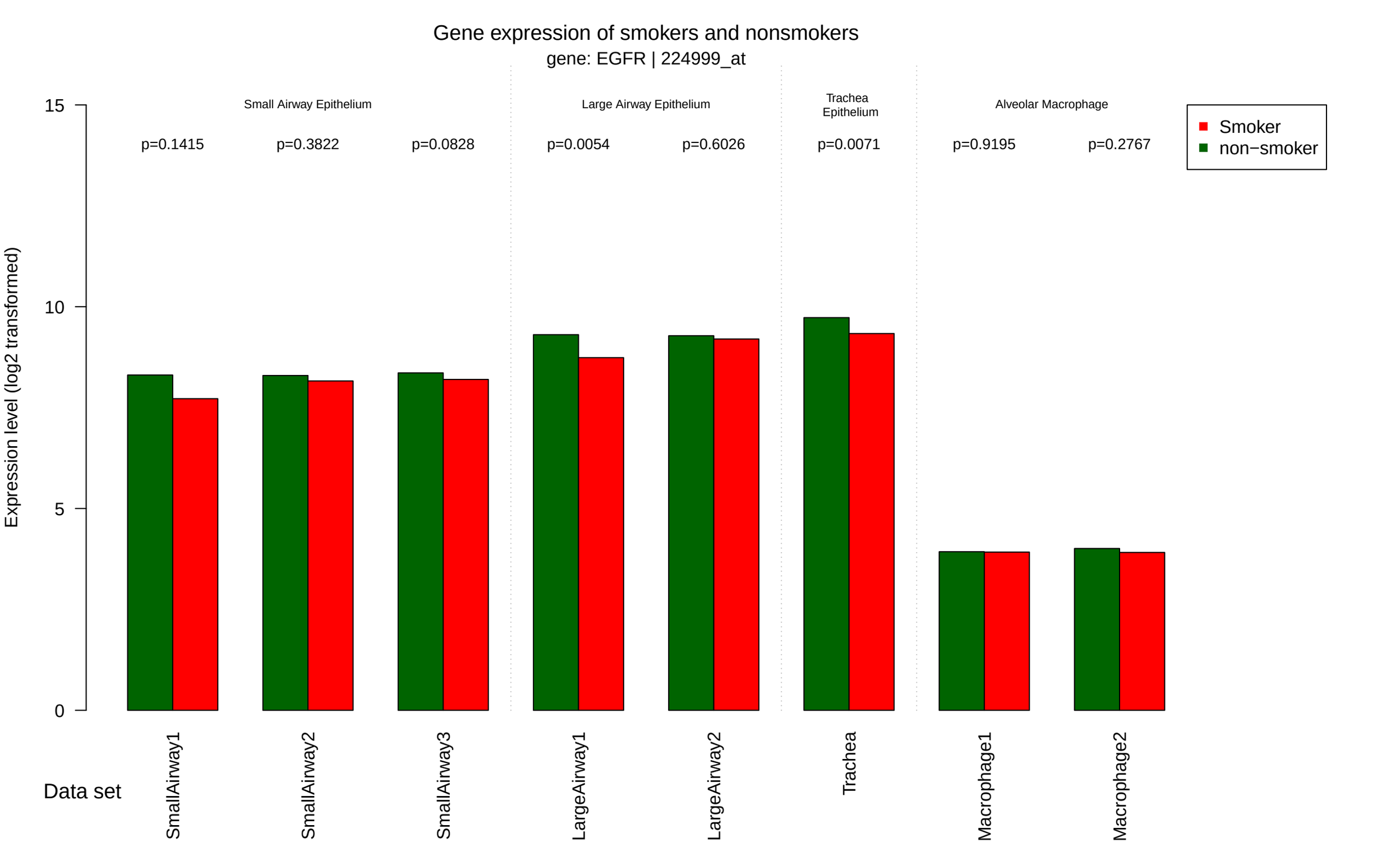
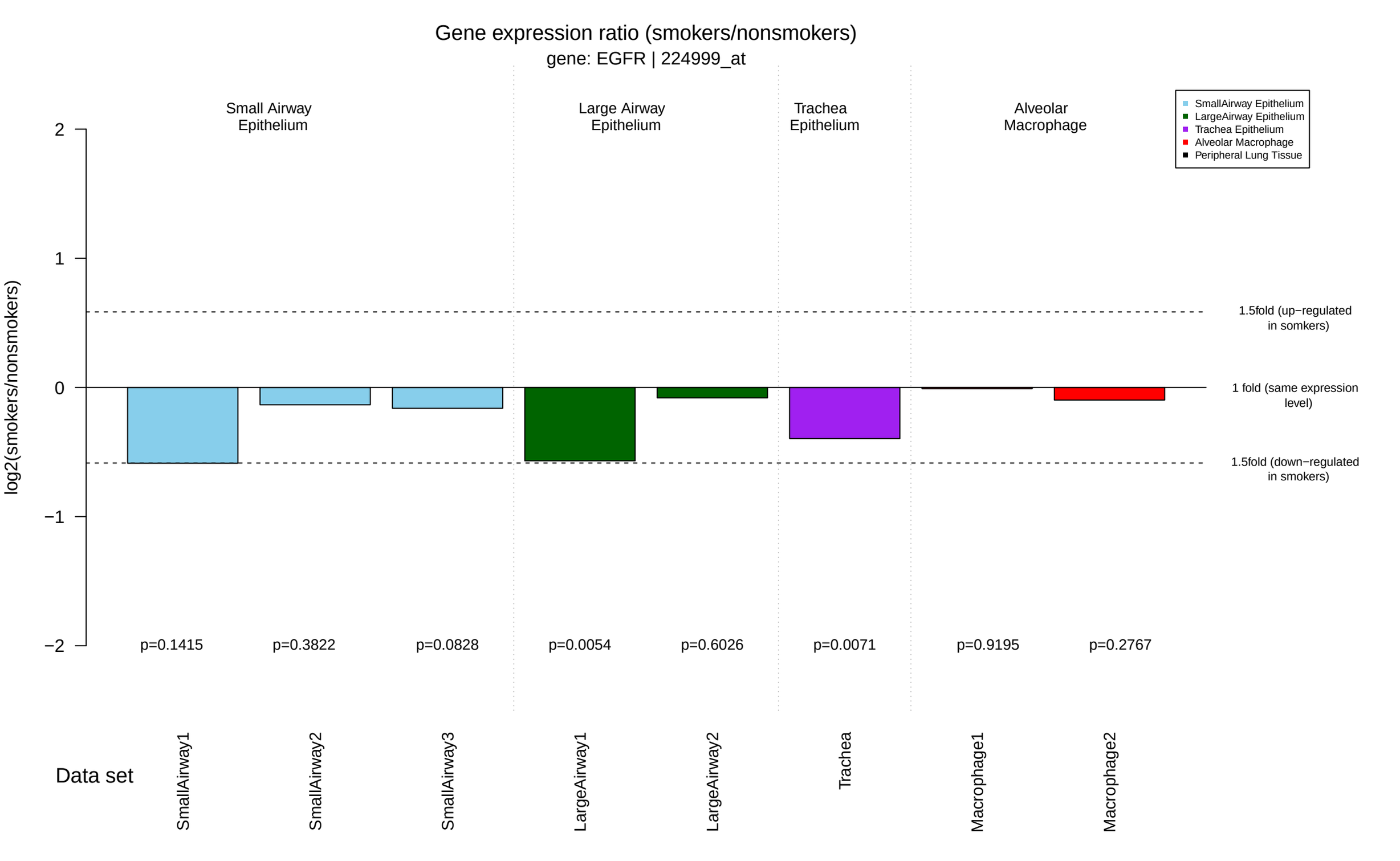
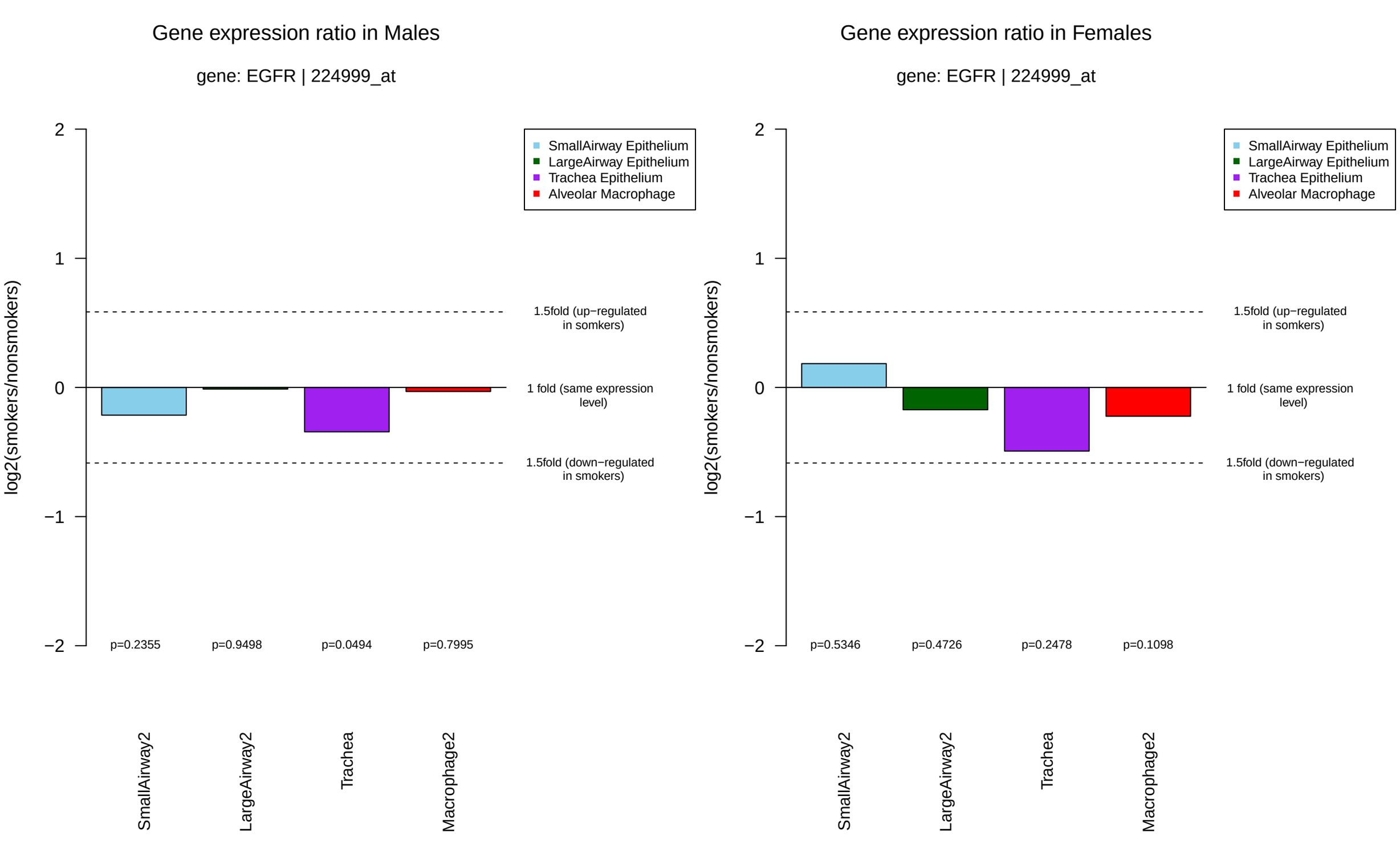
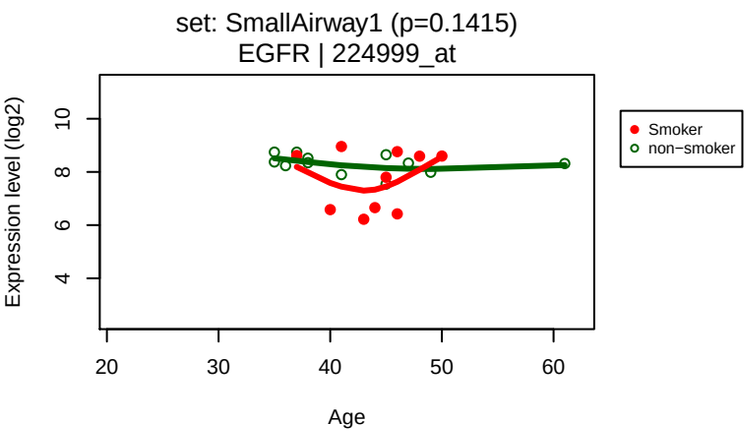
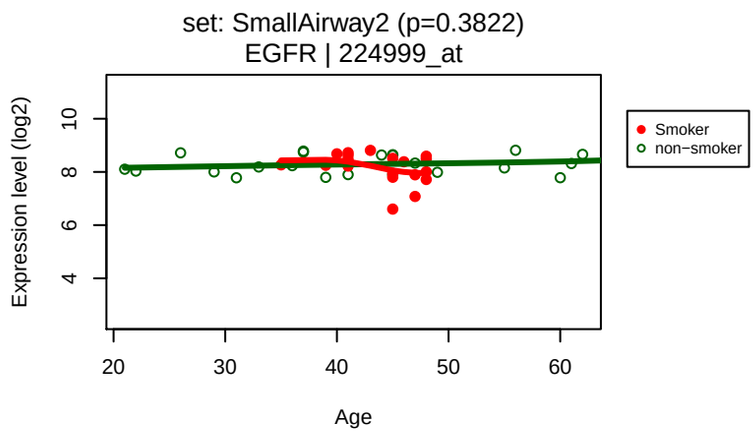
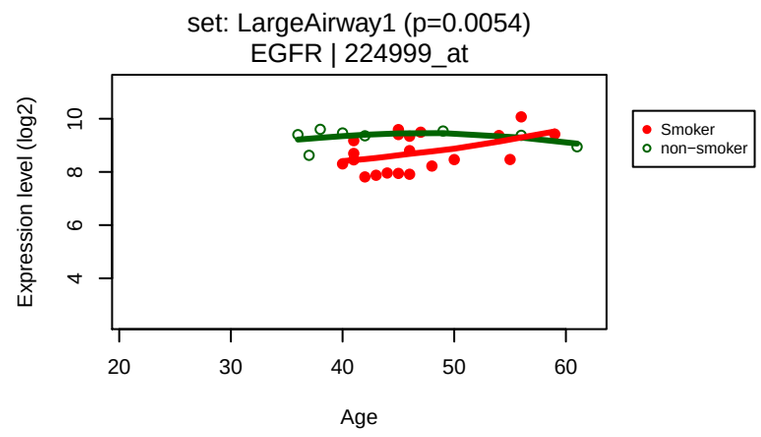

Gene expression of smokers and nonsmokers
gene: EGFR | 224999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1415
p=0.3822
p=0.0828
p=0.0054
p=0.6026
p=0.0071
p=0.9195
p=0.2767
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

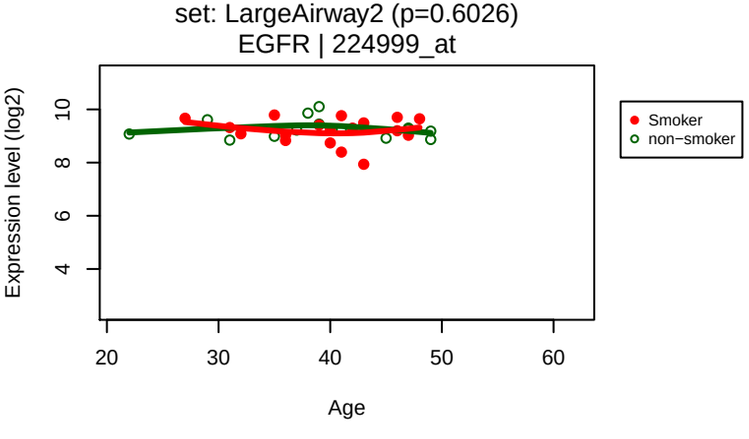

Gene expression ratio (smokers/nonsmokers)
gene: EGFR | 224999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1415
p=0.3822
p=0.0828
p=0.0054
p=0.6026
p=0.0071
p=0.9195
p=0.2767
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

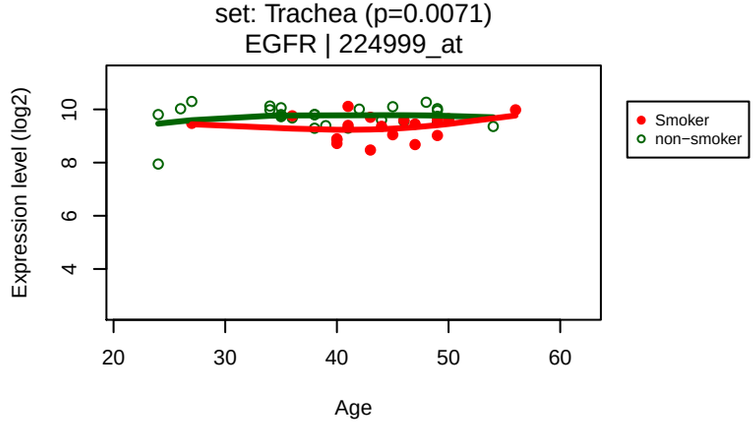

Gene expression ratio in Males
gene: EGFR | 224999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2355
p=0.9498
p=0.0494
p=0.7995
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EGFR | 224999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5346
p=0.4726
p=0.2478
p=0.1098
SmallAirway2
LargeAirway2
Trachea
Macrophage2

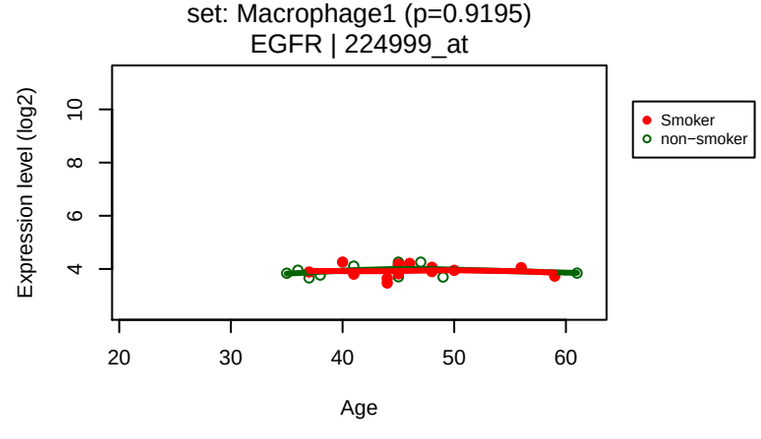

set: SmallAirway1 (p=0.1415)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker

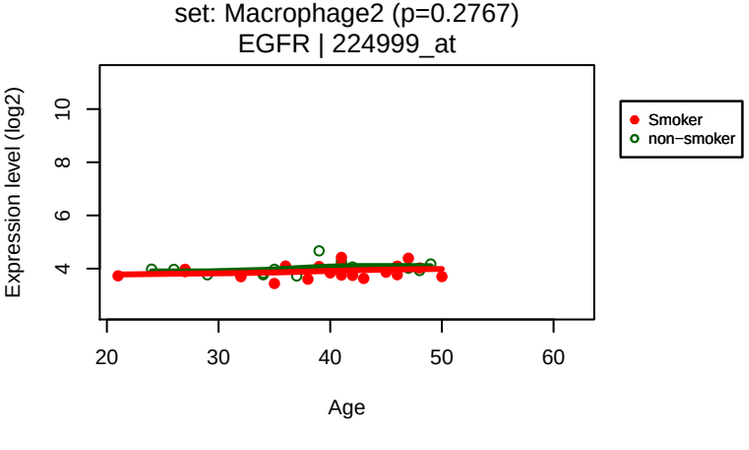

set: SmallAirway2 (p=0.3822)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker

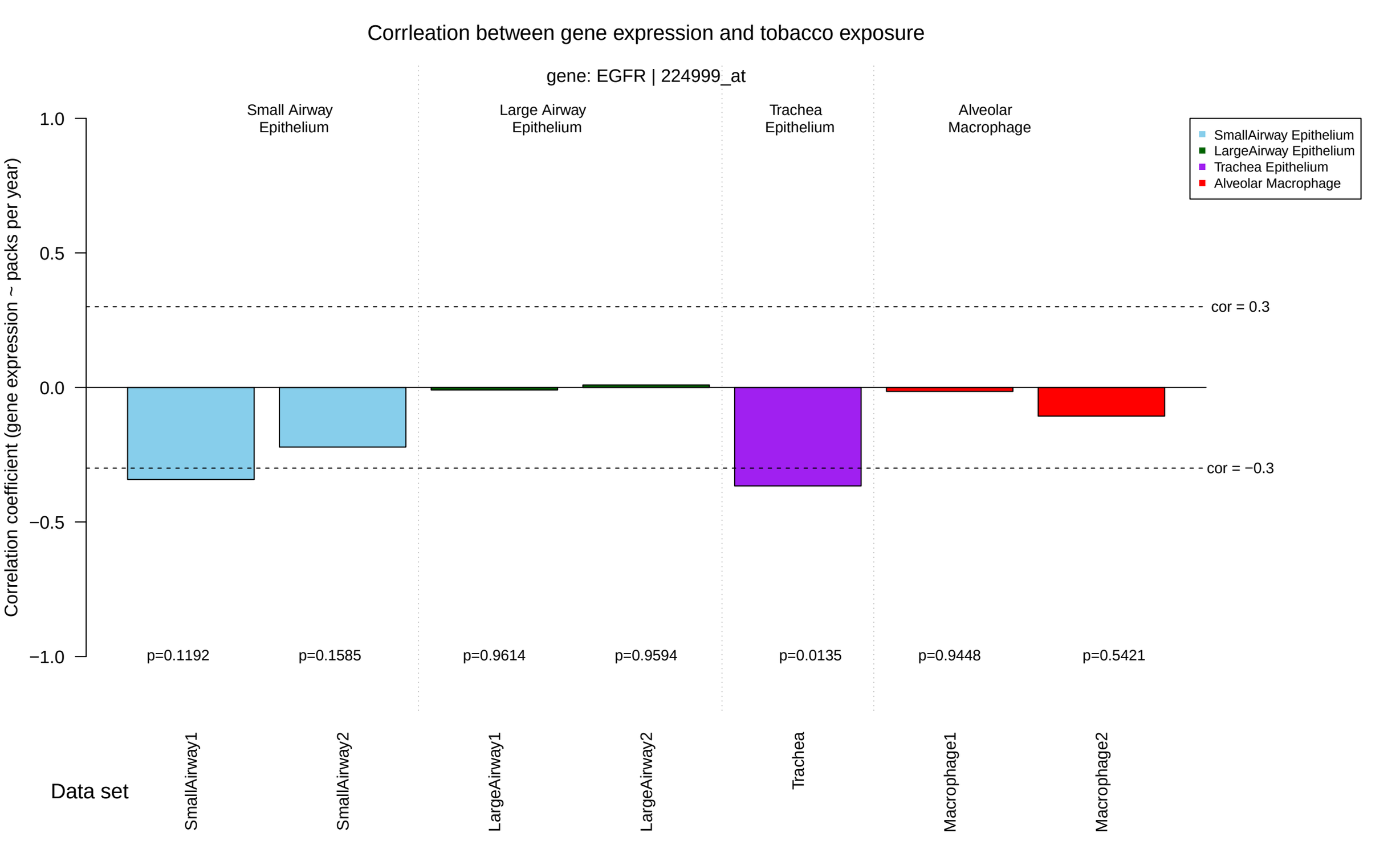

set: LargeAirway1 (p=0.0054)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6026)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0071)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9195)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2767)
EGFR | 224999_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EGFR | 224999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1192
p=0.1585
p=0.9614
p=0.9594
p=0.0135
p=0.9448
p=0.5421
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2